

Technical Monograph on the Implementation of VEGA RISC-V Processors on Genesys 2 FPGA Architectures for Accelerated Neural Network Image Processing

The evolution of semiconductor design has increasingly leaned toward the democratization of hardware through open-standard instruction set architectures (ISAs), with RISC-V emerging as the preeminent framework for custom processor development. This shift is particularly evident in the strategic initiatives of the Government of India, specifically the Digital India RISC-V (DIR-V) program, which has fostered the creation of the VEGA family of microprocessors.¹ These processors, ranging from low-power 32-bit cores to high-performance 64-bit multi-core superscalar out-of-order designs, represent a significant advancement in indigenous computing capability.² Simultaneously, the implementation of such sophisticated silicon IP on field-programmable gate arrays (FPGAs) like the Digilent Genesys 2 provides a robust platform for evaluating performance in demanding domains such as neural network-based image processing.⁴ The Kintex-7 FPGA, which serves as the heart of the Genesys 2, offers a balance of logic density and digital signal processing (DSP) resources that are essential for the high-throughput arithmetic required by modern convolutional neural networks.⁶

The Genesys 2 Development Platform: Architectural Synthesis

The Genesys 2 board is an advanced digital circuit development platform built around the Xilinx Kintex-7 FPGA, specifically the XC7K325T-2FFG900C variant.⁴ This device is categorized as a high-performance FPGA that bridges the gap between cost-optimized and high-end families, making it suitable for video and data processing applications that require substantial logic and memory resources.⁷ The board's design philosophy emphasizes high-speed connectivity and rich peripheral integration, providing the necessary I/O for complex image processing pipelines.

Kintex-7 FPGA Resource Profile

The internal architecture of the Kintex-7 XC7K325T is a critical determinant of the performance of soft-core processors and hardware accelerators. It features 50,950 logic slices, representing a seven-fold increase in capacity compared to previous generations of the Genesys platform.⁶ Each logic slice contains four 6-input Look-Up Tables (LUTs) and eight flip-flops, providing a versatile fabric for the implementation of the complex control logic, instruction decoders, and

execution units found in the VEGA series.⁴

Resource Component	Specification	Functional Relevance in NN/Image Processing
Logic Slices	50,950	Provides fabric for VEGA cores and custom logic ⁴
6-input LUTs	203,800	Implements Boolean logic and small memory structures ⁴
Flip-flops	407,600	Serves as the primary register resource for pipelining ⁴
Block RAM (BRAM)	16 Mbits	High-speed local storage for weights and frame buffers ⁴
DSP48E1 Slices	840	Dedicated hardware for high-speed multiply-accumulate ⁴
GTX Transceivers	16 (up to 10.3125 Gbps)	High-speed data ingestion from camera sensors ⁴
Clock Management	10 CMTs (PLL/MMCM)	Ensures precise timing across various clock domains ⁴

The inclusion of 840 DSP slices is particularly significant for neural network implementation. These slices are optimized to perform complex arithmetic operations, such as $P = (A + D) \times B + C$, in a single clock cycle.⁴ In the context of convolutional neural networks (CNNs), these DSP blocks are the primary engine for performing the sliding-window convolutions that dominate the computational budget. Furthermore, the 16 Mbits of block RAM (BRAM) allows for the implementation of efficient cache hierarchies and local line buffers, which are essential for mitigating the latency of external memory accesses.⁶

Peripheral and Multimedia Connectivity

For image processing, the Genesys 2 is equipped with an array of high-speed video interfaces. These include dual four-lane DisplayPort connectors, an HDMI Sink and Source, and a 16-bit VGA connector.⁴ These ports enable the board to handle real-time video streams, either as a receiver from high-resolution cameras or as a transmitter to high-definition displays. The board also features a 10/100/1000 Ethernet PHY, which is vital for deploying image processing results in networked environments or for receiving training data from remote servers.⁴ Memory bandwidth is supported by 1 GB of on-board DDR3 memory, operating at an 1800 MT/s data rate with a 32-bit data width.⁴ This provides the necessary throughput for large-scale image frames and the vast weight matrices associated with deep neural networks. Expansion capabilities are further extended by a fully-populated 400-pin FMC High Pin Count (HPC) connector and five Pmod ports, allowing for the integration of custom sensors and specialized hardware modules.⁴

VEGA RISC-V Processor Series: Taxonomy and Capabilities

The VEGA series, developed by the Centre for Development of Advanced Computing (C-DAC), is a versatile family of microprocessors based on the open-source RISC-V ISA.² The design goals of the VEGA series include providing indigenous, secure, and scalable computing solutions for a wide range of applications, from low-power Internet of Things (IoT) devices to high-performance superscalar systems.¹

Microarchitectural Classifications

The VEGA processors are structured into several distinct classes, each optimized for different performance and power envelopes. This modularity allows designers to select the most appropriate core for the specific image processing or neural network task at hand.²

Processor Class	Core Model	Architecture Details	Target Application Domain
ET Series	VEGA ET1031	32-bit, single-core, 3-stage in-order ¹¹	Low-power IoT, sensor fusion, smart meters ¹²
ET Series	VEGA ET1037	32-bit CPU IP core based on RISC-V ²	Microcontrollers and simple embedded control
AT Series	VEGA AT1051	32-bit, single-core, 5-stage in-order ²	Industrial automation, mid-range embedded ¹⁰
AS Series	VEGA AS1061	64-bit, single-core, 6-stage in-order ¹⁰	High-performance embedded, edge computing
AS Series	VEGA AS1161	64-bit, single-core, 16-stage Out-of-Order ²	High-throughput data processing, Linux systems
AS Series	VEGA AS2161	64-bit, dual-core, 16-stage Out-of-Order ³	Multiprocessing, complex automation, DHRUV64 ³
AS Series	VEGA AS4161	64-bit, quad-core, 16-stage Out-of-Order ²	Strategic and server-class high-performance computing

The high-end "AS" series processors, such as the AS2161 (DHRUV64), represent a significant microarchitectural achievement. They feature a superscalar out-of-order execution pipeline, typically ranging from 13 to 16 stages.³ This allows the processor to execute multiple

instructions in parallel by analyzing data dependencies and issuing instructions to available execution units, which is particularly effective for algorithms with inherent instruction-level parallelism, such as image filters and feature extraction routines. These cores include full Memory Management Units (MMUs) supporting page-based virtual memory, multi-level caches (L1 and L2), and IEEE 754-2008 compliant floating-point units for single- and double-precision operations.³

RISC-V ISA Foundations and Standard Extensions

The VEGA processors utilize the RISC-V ISA, which is a modular architecture consisting of a base integer set and various optional extensions.¹³ For image processing and neural network workloads, specific extensions are critical for performance.

- **M Extension (Integer Multiplication and Division):** This is foundational for the arithmetic-heavy nature of neural network layers. The inclusion of hardware multipliers allows for rapid execution of the dot products that characterize both convolutional and fully connected layers.¹²
- **A Extension (Atomic Instructions):** Vital for multi-core variants like the AS2161, enabling safe synchronization between cores when processing shared image data or updating neural network parameters in a multi-threaded environment.⁵
- **F and D Extensions (Floating Point):** Image processing often requires high precision for color transformations or complex filtering. The F and D extensions provide single- and double-precision floating-point arithmetic, ensuring accuracy that might be lost in fixed-point implementations.³
- **V Extension (Vector Operations):** Although under active development and profiling for AI workloads, the RISC-V Vector (RVV) extension is the primary mechanism for data-parallel acceleration. RVV 1.0 supports vector-length agnostic (VLA) programming, allowing software to run portably across different hardware with varying vector register widths.¹⁴
- **C Extension (Compressed Instructions):** By reducing the size of common instructions to 16 bits, the C extension improves instruction cache hit rates and reduces memory bandwidth requirements, which is beneficial when the system is under heavy load from data-intensive image processing tasks.¹³

Implementation Strategy on Genesys 2

The deployment of a VEGA processor on the Genesys 2 board involves a multi-stage integration process that maps the RTL (Register Transfer Level) design of the processor and its associated system-on-chip (SoC) components to the FPGA fabric.⁵ This process utilizes the Xilinx Vivado Design Suite for synthesis, placement, and routing, while the software stack is built using the VEGA Software Development Kit (SDK).⁴

SoC Architecture and ASTRA IP Integration

The VEGA cores are typically integrated into a comprehensive SoC, such as the Thejas64, which

was specifically demonstrated as a single-board computer running on the Genesys 2.³ To facilitate rapid SoC development, C-DAC provides a suite of silicon-proven System and Peripheral IP cores under the brand name ASTRA.¹⁰ These IPs are synthesizable, technology-independent, and fully verified, making them ideal for FPGA-based implementation.¹⁹

ASTRA IP Component	Function and Integration Relevance
ERPLIC	Platform Level Interrupt Controller; manages up to 127 interrupt lines from peripherals. ³
ERDMA	Direct Memory Access Controller; offloads the CPU during high-speed transfers of image frames. ¹⁰
ERGMAC	Gigabit Ethernet MAC; enables high-speed networking for image deployment and control. ¹⁰
ERSMC	Static Memory Controller; manages access to SRAM and NOR flash for bootloading. ¹⁰
ERSDHOST	SD Host Controller; allows the system to boot and load data from Micro SD cards. ¹⁹
ERUSB2	USB 2.0 Host/Device/OTG; provides connectivity for human interface devices or storage. ⁴
ER16C450	UART Controller; serves as the primary console for debugging and system interaction. ³

The integration on the Genesys 2 involves connecting the VEGA core to these ASTRA IPs using high-performance interconnects, such as AXI4 (Advanced eXtensible Interface) or ACE (AXI Coherency Extensions).³ This interconnection fabric manages the data flow between the processor, memory controllers, and peripheral devices. The Thejas64 SoC on Genesys 2 has been demonstrated booting a full Linux kernel (version 4.18) using the Buildroot build system, complete with the X11 windowing system and Fluxbox window manager, showcasing its capability as a personal computer based on indigenous technology.⁵

Hardware Emulation and Debugging Workflow

The Genesys 2 board serves as an essential hardware emulation platform in the development lifecycle of the VEGA processors.² Before committing to the high costs of ASIC fabrication at facilities like the Semiconductor Lab in Mohali, the FPGA implementation allows for exhaustive verification of the RTL in a real-world operating environment.³

The programming and debugging workflow is facilitated by the VEGA SDK, which is an Eclipse-based environment that includes:

1. **GNU-based Toolchain:** A cross-compiler, linker, and assembler for the RISC-V ISA, allowing developers to target the specific extensions present in the VEGA core.³

2. **Board Support Package (BSP):** A collection of bare-metal device drivers and initialization files for the ASTRA peripherals, enabling low-level control of I2C, UART, SPI, and PWM modules.¹⁸
3. **JTAG Debugging:** The Genesys 2's on-board USB-JTAG programmer allows for real-time debugging using OpenOCD and GDB. The VEGA cores include integrated debug controllers that support hardware and software breakpoints, enabling developers to step through code and inspect registers directly on the FPGA.³
4. **Emulation Support:** The ecosystem supports architectural simulators and emulators like QEMU and Spike, providing a software-only path for initial development before moving to the Genesys 2 hardware.¹⁸

Neural Network and Image Processing Acceleration Mechanisms

The primary motivation for offloading image processing and neural network inference to an FPGA-based RISC-V system is the inherent parallelism of these tasks, which often overwhelms general-purpose CPUs.²² In a coupled VEGA and Genesys 2 system, acceleration is achieved through a combination of microarchitectural features, custom instructions, and dedicated hardware accelerators.

Dedicated Hardware Accelerators and IP Cores

A common approach for high-performance neural network inference is the instantiation of dedicated IP cores within the FPGA fabric that work alongside the VEGA processor. These cores are designed to accelerate the fundamental operations of convolutional neural networks.²⁴

- **Convolution Operation IP Cores:** These cores implement the multidimensional sliding-window MAC operations that form the core of CNN layers. By utilizing the Kintex-7's DSP48E1 slices, these cores can perform hundreds of multiplications in parallel, significantly outperforming a scalar CPU core.⁴
- **Pooling Operation IP Cores:** These units handle the spatial reduction of feature maps, typically through Max or Average pooling, which reduces the computational load for subsequent layers.²⁴
- **Full-Connection Operation IP Cores:** Optimized for dense matrix-vector multiplications, these cores are essential for the final classification layers of a neural network.²⁴
- **Non-linear Activation Units:** Hardware implementations of functions like ReLU (Rectified Linear Unit), Sigmoid, and Softmax are integrated into the pipeline. These often use look-up tables (LUTs) or piecewise linear approximations to maintain high throughput without the overhead of complex floating-point calculations.¹⁶

These accelerators are often integrated into the SoC via specialized interfaces. One such interface is the Extension Acceleration Interface (EAI), which connects a general-purpose accelerated coprocessor directly to the RISC-V CPU core.²⁶ This allows the main VEGA core to

dispatch intensive tasks, such as object detection or bilateral filtering, to the accelerator while maintaining overall system control.²⁶

Custom Instruction Set Extensions

The modularity of the RISC-V ISA allows for the definition of custom instructions that are unique to a specific image processing algorithm.²³ By adding custom execution units to the VEGA processor's pipeline, a single RISC-V instruction can replace a lengthy software subroutine.

For example, a custom instruction for bilateral filtering—a common technique for image denoising—can be implemented to calculate the spatial and intensity weights in hardware. This drastically reduces the instruction count and memory traffic compared to a pure software implementation.²⁷ Documentation of similar custom hardware optimizations has shown speed improvements as high as 23x over software-only computations, transforming a 23-second task into a 1-second task.²⁸ This capability allows the VEGA processor to be transformed into a Domain-Specific Architecture (DSA) tailored for edge AI applications.²³

Image Processing Pipeline Implementation

The Genesys 2 and VEGA system is capable of hosting comprehensive image processing pipelines. These pipelines typically involve:

1. **Ingestion:** High-speed raw image data is received via DisplayPort, HDMI, or the FMC connector.⁴
2. **Preprocessing:** Spatial filters such as Sobel for edge detection, or Gaussian blurs for noise reduction, are applied using either the VEGA core's vector extensions or dedicated DSP-based hardware.²⁹
3. **Feature Extraction:** Convolutional layers of a neural network (e.g., YOLOv3 or MobileNet) are executed to identify key patterns and objects within the frame.³⁰
4. **Classification/Detection:** Fully connected layers and activation functions generate the final output, such as object coordinates or class probabilities.²⁴
5. **Output Display:** The processed image, with overlays or bounding boxes, is transmitted to a monitor via the VGA or HDMI Source ports.⁴

This system has been evaluated for resource utilization and accuracy on various platforms, demonstrating its robustness in computationally intensive tasks like driver drowsiness detection using trained CNNs.²⁷

Implementation Bottlenecks and Reasoning

Despite the significant advantages of the Genesys 2 and VEGA platform, the implementation of complex neural networks on FPGA-based soft processors faces several technical bottlenecks. Understanding these constraints is essential for effective system design and optimization.

The Von Neumann Bottleneck and Memory Bandwidth

One of the most persistent challenges in high-performance computing is the von Neumann

bottleneck, defined by the limited bandwidth and high latency of the data path between the processing unit and the main memory.³² Neural networks, particularly deep CNNs, require the transfer of millions of weights and massive feature maps for every frame processed. On the Genesys 2, while the DDR3 memory provides 1 GB of capacity, the physical interface and the AXI interconnect hierarchy can become saturated.³ If the VEGA processor or the hardware accelerator must constantly wait for data from external memory, the high-speed execution units will remain idle, leading to a "memory-bound" system. FPGAs attempt to mitigate this by using internal block RAM (BRAM) to collocate memory and processing.³² However, with only 16 Mbits of BRAM available on the Kintex-7, it is often impossible to store entire model weights on-chip, necessitating complex memory management techniques like weight tiling and data compression.⁶

Logic Density and Timing Closure

Implementing a sophisticated 16-stage out-of-order processor like the VEGA AS1161 on an FPGA requires significant logic resources.² As the design approaches the limits of the 50,950 logic slices, the FPGA's routing fabric becomes congested.⁴

- **Congestion and Timing:** High congestion leads to longer routing paths between logic elements, which increases signal propagation delay. This makes it difficult to achieve "timing closure" at high clock frequencies. While the Kintex-7 is capable of clock speeds exceeding 450 MHz for simple designs, a complex VEGA-based SoC may be forced to run at lower speeds (e.g., 50-150 MHz) to maintain stable operation.⁶
- **Resource Trade-offs:** Designers must often choose between features and speed. For instance, including a full double-precision floating-point unit (D extension) consumes a large number of DSP slices and LUTs, potentially leaving fewer resources for a dedicated neural network accelerator or requiring a reduction in the number of parallel processing elements.²⁷

Toolchain and Software Complexity

The flexibility of the RISC-V architecture also introduces complexity in the software development process. When custom instructions or extensions are added to a VEGA core to accelerate image processing, the standard toolchains (compilers like GCC or LLVM) must be modified to recognize these new opcodes.¹⁵

- **Compiler Optimization:** Standard compilers are often unoptimized for custom hardware extensions. To achieve the best performance, developers may need to write hand-tuned assembly kernels or use C intrinsics to explicitly invoke specialized hardware units.¹⁵
- **Vector Programming:** The shift toward vector length agnosticism (VLA) in the RVV 1.0 specification requires a change in programming mindset compared to traditional fixed-width SIMD (Single Instruction, Multiple Data) architectures like ARM NEON. While VLA ensures long-term portability, it increases the complexity of the compiler's scheduling and register allocation logic.¹⁵

Strategic Implementation Reasoning

The decision to use an FPGA-based RISC-V system like the Genesys 2 and VEGA coupling is driven by several critical factors:

1. **Reconfigurability:** Unlike an ASIC, which is fixed once manufactured, the FPGA can be reconfigured to support different neural network architectures (e.g., switching from a CNN for object detection to a Vision Transformer for image segmentation) without changing the hardware.²²
2. **Cost and Prototyping:** For researchers and companies in the early stages of development, the Genesys 2 board provides a cost-effective way to validate indigenous processor designs without the multi-million dollar expense and multi-month lead times of silicon tape-outs.²²
3. **Security and Sovereignty:** The use of an indigenous processor family like VEGA is vital for strategic applications (defense, aerospace, critical infrastructure) where trust in the underlying silicon is paramount. An open-source ISA like RISC-V allows for full transparency and verification of the processor's RTL.²

Comparative Performance Analysis and Advantages

The performance of image processing on a VEGA-accelerated Genesys 2 system can be evaluated through benchmarks and comparisons with traditional computing architectures.

Performance vs. Traditional CPUs and GPUs

While general-purpose CPUs (GP-CPU) are flexible, they lack the massive parallelism required for high-resolution, high-frame-rate video processing.²² GPUs offer excellent parallelism but are often power-hungry and over-designed for simple embedded tasks.

Platform	Relative Power Consumption	Reconfigurability	Performance/Watt (NN Tasks)	Best Use Case
GP-CPU	High	None (Fixed)	Low	General data processing ²³
GP-GPU	Very High	None (Fixed)	Moderate	Big-data analytics, CNN training ²³
FPGA (Kintex-7)	Moderate	High	High	Edge AI, real-time image processing ²²
Custom ASIC	Very Low	None (Fixed)	Extremely High	Mass-market consumer devices ²²

FPGAs demonstrate "very high performance" by fully extracting the parallelism inherent in image processing applications, despite their lower operational frequency compared to modern high-end CPUs.²² For example, in signal conditioning tasks, a RISC-V system with a dedicated

signal processing extension has been shown to perform 4.52x faster than standard MicroBlaze or CVA6 cores.³⁵ In real-time laser speckle imaging, a dedicated hardware processor in an FPGA running at 130 MHz can process up to 85 raw images (640x480 resolution) per second, a feat that would be difficult for a traditional embedded CPU to achieve with the same power budget.³³

Advantages of the VEGA and Genesys 2 Ecosystem

1. **High-Efficiency Parallelism:** The combination of VEGA's superscalar architecture and the Kintex-7's DSP slices allows for the concurrent processing of image data, which is essential for low-latency applications like autonomous robot control or real-time digital image reconstruction.³²
2. **Modular Scalability:** The VEGA series' range (from 32-bit ET to 64-bit multi-core AS) allows for a "right-sized" implementation. A low-power sensor fusion task might use an ET1031, while a complex object detection system might utilize an AS2161 with a dedicated NPU coprocessor.¹²
3. **Indigenous Innovation and Ecosystem:** The Digital India RISC-V program has not only produced the processors but also the complete ecosystem of ASTRA IPs, SDKs, and development boards, enabling fully indigenous product development for strategic sectors.²
4. **License-Free Flexibility:** By leveraging the RISC-V ISA, C-DAC and its partners can innovate without paying the significant licensing fees associated with proprietary architectures like ARM. This freedom allows for the addition of custom extensions that provide a competitive edge in specialized domains like AI and crypto-acceleration.¹⁵

Future Trends and Strategic Outlook

The landscape of AI acceleration on RISC-V is rapidly shifting from primarily vector-oriented computation toward large-scale matrix operations, driven by the rise of Large Language Models (LLMs) and advanced Vision Transformers.¹⁶

The Shift to Matrix Extensions

Workloads in modern AI kernels spend substantial time in activation and normalization functions (LayerNorm, Softmax, Sigmoid), which can constrain throughput even when the main linear algebra is accelerated.¹⁶ Future iterations of the VEGA processors are likely to explore:

- **Reduced-Precision Data Types:** Broader support for BF16 (BFloat16) and FP8 formats is being standardized through efficient encoding mechanisms, often involving a single-bit toggle in the vtype control register to avoid long instructions.¹⁶
- **Independent Matrix State:** New extension proposals define independent matrix registers to handle operands and results, offering broad design flexibility at the cost of increased load/store complexity.¹⁶
- **Dedicated AI Platforms:** The combination of RVV 1.0 with a dedicated Neural Processing Unit (NPU) provides the basis for an AI-native compute platform, capable of replacing

fixed NPUs in specific segments like intelligent IoT edge devices.¹⁶

Strengthening the Indian Ecosystem

The successful porting of the VEGA processors to the Genesys 2 board is a milestone for the Microprocessor Development Programme sponsored by MeitY.³ With related initiatives like SHAKTI (IIT Madras) for defense, AJIT (IIT Bombay) for industrial robotics, and VIKRAM (ISRO-SCL) for space missions, India is establishing a robust and independent semiconductor roadmap.³

In conclusion, the coupling of the Genesys 2 board and VEGA processors offers a powerful, flexible, and efficient platform for accelerating neural network-based image processing. By navigating the technical bottlenecks of memory bandwidth and logic density through sophisticated SoC design and custom hardware acceleration, this indigenous ecosystem provides a viable and strategic alternative to proprietary silicon. As the RISC-V architecture continues to evolve with stable vector and matrix extensions, the VEGA series is well-positioned to drive innovation in high-performance embedded and edge AI applications.

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